

Winning and Attendance in the WNBA

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This paper examines how the relationship between team quality, game quality, and attendance has changed over the history of the WNBA. In general, the literature expects that stronger teams will see higher live game attendance in professional sports. However, previous research and analyst commentary has cast doubt on the strength of the relationship between team strength and game attendance in the WNBA. Our study improves upon previous research by studying attendance in the broader context of the league’s growth during 1997-2025, finding that the factors that drive fan demand have evolved as the league has matured. We test whether three leading theories of fan behavior- quality of play, loss aversion, and outcome uncertainty- have become more important drivers of attendance as the league has grown. We use game-level attendance data reported directly by the WNBA from the “Across the Timeline: WNBA League Attendance Dashboard” as well as team ELO ratings and game-by-game home win probabilities and find that in recent years fans are increasingly likely to attend games in response to these factors. ELO ratings and win probabilities offer a significant advantage to other measures of team strength or outcome likelihood because they are dynamic and incorporate information available up to the date of each game. Our results suggest that the WNBA is entering a new era in which fans are choosing to attend games to watch good basketball rather than novelty-driven interest or broader support for women’s sports.

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1 Introduction

In the years prior to the 2020 WNBA collective bargaining agreement (CBA), NBA Commissioner Adam Silver publicly characterized the women’s league as unpopular and unprofitable (Garcia 2020). Again during the CBA renegotiations in 2025, NBA and WNBA leadership

made similar negative claims about the women’s league’s long term viability. While league leadership has been ironically pessimistic about the health of the WNBA in recent years, their claims stand in tension with recent trends: the league has enjoyed a significant increase in attendance and fan interest in the past several seasons.

While this trend is encouraging for anyone interested in the success of the WNBA, breaking down further the underlying components of this trend is necessary both to explain and sustain this growth. For decades, league officials and sports economists have debated the primary drivers of game attendance, identifying game-related and market-related factors impacting fan interest and game attendance. Much of this research, beginning with foundational work in the 1970s, assumes that fans are more likely to attend games when the home team is expected to do well. This relationship is widely tested and supported in men’s professional sports, but women’s leagues and the WNBA in particular are still largely underresearched. As a relatively young league with rapid growth in viewership (WNBA 2024) but limited publicly available financial data, game attendance remains the most readily available proxy variable for public interest in the WNBA. In this paper we set out to determine how the underlying causes shaping game attendance work to explain these more recent trends, and specifically whether on-court factors are becoming more important to WNBA game attendance in recent years.

1.1 Existing scholarship on team performance and game attendance

A key line of research in sports economics links fan demand to expectations about game outcomes. Early work such as the uncertainty-of-outcome hypothesis suggests that attendance is higher when game results are less predictable, increasing perceived excitement. In their 2014 paper, Coates et al. use more recent behavioral economics models to suggest that people evaluate gains and losses relative to a reference point and tend to dislike losses more than they like equivalent wins (Coates et al. 2014). They argue that applying this idea of loss aversion may better explain why fans choose to go games than the uncertainty of outcome hypothesis. They use MLB live-game attendance data to find attendance patterns consistent with the hypothesis that fans are less likely to attend games in which they expect the home-team to lose (Coates et al. 2014). These frameworks together suggest that attendance decisions are sensitive not only to team quality, but to expectations about competitive performance and likely game outcomes.

In the first study in sports economics that sought to empirically identify the factors that affected attendance, Noll (1974) considered data from MLB, the NFL, NHL, NBA and ABA (American Basketball Association) for the 1969-70 and 1970-71 seasons and found that demand was a function of population, interactions of income with star players, team win percentage, and per capita income. Multiple subsequent studies on men’s professional team sports continued to find a statistically significant correlation between team quality and attendance.

Differences in the literature over how to measure team quality (wins, wins lagged, number of championships in past years, win probability, star power), attendance (logs, lags, linear regression or censored), the inclusion of control variables and non-linear forms also emerged

(Wiseman and Chatterjee 2003; Woltring et al. 2018; Kim and Chang 2023; DeSchriver and Jensen 2002; Paul et al. 2019).

The relationship between attendance and winning percentage has been examined, but such studies often look at winning percentage as causation for attendance. Davis (2008) and Horowitz (2007) both found that attendance and winning percentage were related. Horowitz concluded a bidirectional relationship between the two, while Davis suggested that causation runs from winning percentage to attendance. In a following study, Davis (2009) found winning percentage as a significant determinant of attendance for 12 National League teams; though the study did not establish causation both ways. Lemke et al. (2010) found attendance increases as home team win probability increases. There is evidence that causation may run from attendance to winning percentage, but the bulk of the literature has been researched the other way.

Finally, situating these revenue models within the broader league structure, sports economists have written extensively about the relationship between winningness and team revenue, particularly about how competitive balance benefits teams across the league in generating revenue. While teams and fans want to win as many games as possible, significant imbalance in winning percentages across a professional sports league will tend to decrease viewership as spectators will only watch a game if they have some uncertainty about the outcome (Perline et al. 2018). In their study of competitive balance in NCAA basketball, Perline et al. found that the women's game was significantly less balanced than the men's game, looking at the standard deviation and the range of winning percentages for teams in the women's Division I (Perline et al. 2018). They hypothesize that the difference in the competitive balance of the men's and women's games could be one cause of lower revenue generation and attendance in the women's game.

1.2 Why is the WNBA different?

In their study of attendance determinants in the WNBA, Agha and Berri argue that a major reason why attendance is considerably lower at WNBA games than NBA games is that the women's league is still in its early years, only beginning play in 1997 compared to the NBA's 1946. To account for this, they compare attendance in the tenth season of the WNBA (2006) and the NBA (1955/56) (Agha and Berri, 2023, p. 36.) In their respective tenth years, the WNBA average attendance per home game exceeded the NBA's by more than 2,000 fans per game. This comparison suggests that the current disparity in attendance between the women's and men's league should not be interpreted as a lack of demand, but rather a sign of the league's maturity. Considering the league's age as a factor in demand for attendance, it is interesting to inquire further into whether patterns of fan behavior change as the league matures. Agha and Berri attempt to establish whether different forces shape demand for attendance in the WNBA and the NBA, finding that team performance was not a statistically significant predictor of game attendance in the WNBA. As noted above, this is contrary to the virtually unanimous finding in the existing scholarship for men's leagues. However, their study only employed data from 2004 through 2017 and thus can not capture the more recent improvements in attendance shown in figure 1. They also measure a team's strength using the team's winning percentage

from the previous year which raises issues because fans are more likely to respond to the more recent performance of the team. Thus, while a team’s win percentage in 2016 might have an appreciable impact on the team’s attendance at the beginning of the 2017 season, as the season wears on, this impact should consistently diminish. For this reason, we chose to use a team’s ELO rating as the measure of team strength because the ELO rating updates with each game, thus more accurately reflecting the information that fans have access to in deciding to attend games.

Despite the issues with Agha and Berri’s study, their observation that between 2004 and 2017 the relationship between team strength and attendance in the WNBA was weak is not incorrect. The disparity between raw attendance numbers and attendance driving factors in the NBA and WNBA cannot be attributed to the age of the leagues alone. Additional important factors include: the WNBA plays during the summer months in order to promote arena use outside of the NBA season; there is less money spent on promotion; reduced media coverage; historical prejudice; arena amenities and age; and ticket prices, among others. Sport management literature also includes several studies that find a major motivation behind fandom for women’s sports leagues to be an area’s overall support for feminist issues, suggesting that fans in those areas may attend games out of political support rather than to see good basketball (Delia 2020; Guest and Luijten 2018; Leslie-Walker and Mulvenna 2022). Factors like the summer season, reduced media coverage, and the association of game attendance with feminist causes all contribute to the argument that the WNBA functions fundamentally differently to the NBA- at least in terms of attendance drivers. This conception of the WNBA as fundamentally different from the NBA is dangerous both to the league’s reputation and future growth, motivating our more thorough examination of game attendance in the context of the league’s history.

It is reasonable to anticipate that as women’s sports mature and are perceived increasingly as a normal sports league, rather than a political concession or oddity, they will begin to assert similar fan demand characteristics. In particular, fan demand will become more responsive to team quality. Using game attendance¹ and team performance data from 1997 through 2025, we test these hypotheses. In viewing more recent attendance figures for the WNBA in the context of the league’s history, we hoped to determine whether predictors of attendance have shifted over time, due to the WNBA’s relative instability as a young league. Specifically, we find that in recent years the strength of the effect of a game’s uncertainty, the strength of both teams, and the likelihood of a home team win, on game attendance has increased, paralleling documented trends in men’s leagues.

¹Note that, as in most sports leagues, there is always the possibility of finagling with game attendance figures. The WNBA appears to give attendance counting instructions to teams that define attendance as tickets out, including total ticket sales, plus complimentary tickets and promotions. There is considerable room for these factors to be manipulated, rendering the possibility of noise in the attendance figures we use.

2 Data

2.1 Game logs and attendance

Our analysis is based on game-level data that spans WNBA history, beginning in 1997 and continuing through the end of the 2025 season. Our final data set contains observations from 6,164 games.

First, we collected game-level attendance data from (Across the Timeline, n.d.), an online repository for women’s basketball data. The site collects attendance numbers directly from the league’s records. Attendance data was unavailable for only 152 games, while (due to the COVID-19 pandemic) 2 games (accurately) recorded attendance of 0. Because the arenas WNBA teams play in vary widely in size— the New York Liberty alone have called both the Barclays Center (17,732 seat capacity) and the Westchester County Center (2,300 seat capacity) home in recent years— we contextualize raw attendance numbers by approximating the capacity of each game venue. Because accurate records of arena configurations and capacities were difficult to obtain (particularly for venues not still in use) we approximated the practical capacity of each space by pulling the maximally attended game for each venue. While teams have many reasons they might choose to play for a sold-out crowd in a smaller arena, smaller venue size will also by nature constrain attendance, making its consideration in our model important.

Second, we collected game-level performance data from the **wehoop** R package (Gilani and Hutchinson 2021), which uses the official WNBA API. The game-level performance data include These data include the final score of the game. We also used the **wehoop** package to identify whether Caitlin Clark and/or A’ja Wilson played more than 0 minutes in the game.

Table 1 shows a sample of the data. While many other variables are omitted from the table, we note that the attendance, the teams playing in the game, the outcome, the arena, and whether Clark or Wilson played in the game, are recorded.

Source: [Article Notebook](#)

Figure 1 shows the distribution of attendance across all seasons, with a logarithmic scale on the vertical axis. There is no attendance data from 2020, because all games that year were played in the WNBA bubble due to the COVID-19 pandemic, and some games in 2021 continued to be played with artificially low attendance.

Source: [Article Notebook](#)

Several broad observations can be made from Figure 1. First, after some initial enthusiasm during the first three years of the league, attendance largely plateaued or steadily but slightly declined from 2000 to 2017. The two years preceding the pandemic (2018-2019) had notably lower attendance. Since the pandemic, attendance has rebounded, exceeding its initial levels in 2024 and reaching new heights in 2025. In fact, average WNBA attendance in 2025 was 10,986, which was 1.1% higher than the previous high in 1998.

Table 1

WNBA Game Data (1997-2025)

Sample of game-level records

Game Date	Home	Away	Arena	Attendance	Plus Minus	Clark	Wilson
14402	NYL	CON	Madison Square Garden	13,397	-9	0	0
12573	WAS	CON	Verizon Center	13,246	-9	0	0
14461	IND	CHI	Conseco Fieldhouse	6,581	9	0	0
12977	IND	CON	Conseco Fieldhouse	8,799	11	0	0
11134	ORL	HOU	Amway Arena	7,427	-12	0	0
17396	MIN	IND	Xcel Energy Center	9,621	59	0	0
12220	NYL	SAC	Madison Square Garden	10,711	9	0	0
11169	DET	CLE	Palace of Auburn Hills	4,857	-11	0	0
16971	CON	SAN	Mohegan Sun Arena	5,596	3	0	0
16611	SAN	PHO	Freeman Coliseum	3,102	5	0	0

Sample of Game-level WNBA data. The data indicates (among other things) which teams were playing, who won, the arena, the attendance, and whether Caitlin Clark and/or A’ja Wilson played in the game.

These last two years coincide with the arrival of Caitlin Clark, the 2024 Rookie of the Year Award winner who is widely regarded as a generational talent. Disentangling the “Caitlin Clark effect” from the league’s overall attendance trends is one of our goals.

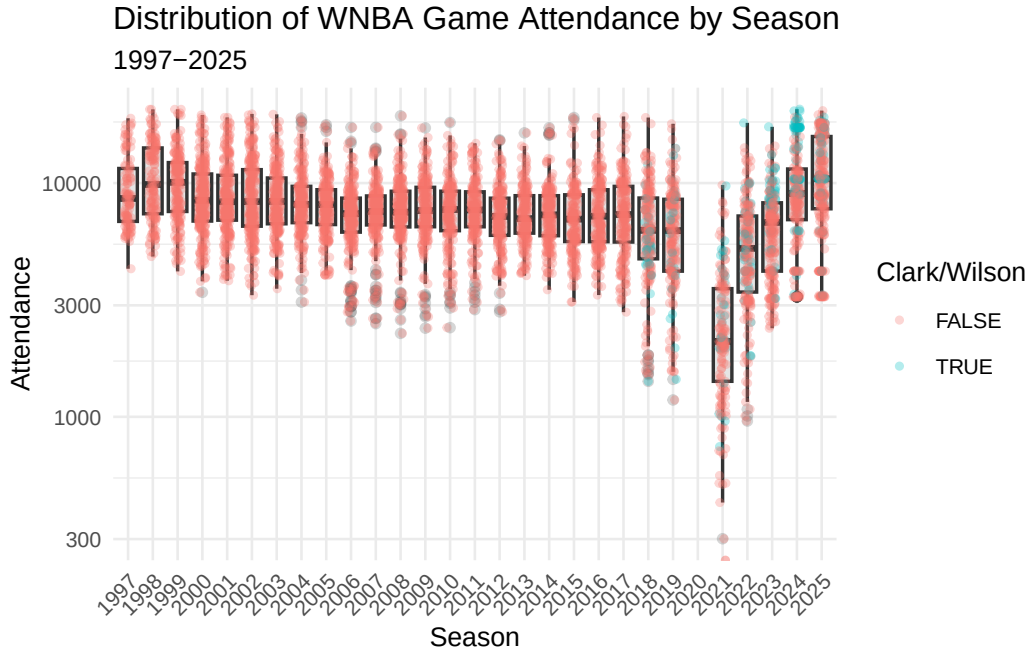


Figure 1: Distribution of WNBA game attendance by season. Note that the vertical axis has a logarithmic scale. Caitlin Clark and/or A’ja Wilson’s games are separated by color.

Source: [Article Notebook](#)

Figure 2 shows the distribution of attendance as a percentage of the arena’s capacity. Although we recorded reported arena capacities, those capacities change over time, and some games had attendance that exceeded the arena’s reported capacity. Instead, we computed the percentage as a ratio of the attendance to the larger of the reported capacity, or the maximum recorded attendance at that arena in the sample.²

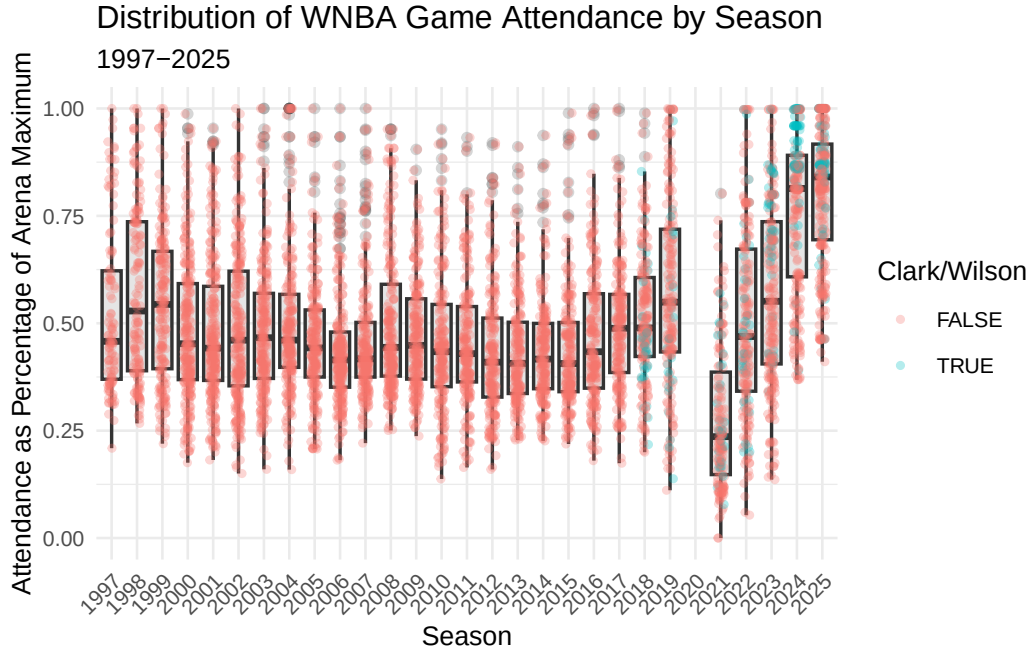


Figure 2: Distribution of WNBA game attendance as a percentage of arena capacity by season. Caitlin Clark’s games in 2024-2025 are separated by color.

Source: [Article Notebook](#)

2.2 Estimating team strength

We used the game-level results data to model the strength of each WNBA team as a time series. Measuring team strength is necessary to test our hypothesis that attendance is in part driven by the desire of fans to see quality basketball. We considered season-to-date winning percentage and a rolling 44-game (the length of a typical WNBA season) winning percentage before settling on Elo rating (Elo 1978). Originally developed for rating chess players, Elo rating is well-understood in sports analytics as a measure of team strength that updates after

²These modifications reduced the number of 1’s in the sample, which was useful mathematically during modeling.

each game, and—unlike winning percentage and rolling winning percentage—takes into account not only the outcome but also the estimated team strength of both teams. This means that beating a team with a higher Elo rating has a greater positive effect than beating a team with a lower Elo rating, reflecting upsets with more nuance than the win-loss record.

Let $\theta_i(t)$ be the strength of team i at time t , where Elo ratings are typically centered at 1500. Then if the probability that team i beats team j is modeled by a logistic curve with base 10 and scale factor 400, the win probability is given by:

$$\Pr(i \text{ beats } j \text{ at time } t) = \frac{1}{1 + 10^{\frac{\theta_j(t) - \theta_i(t)}{400}}}.$$

Each team starts the 1997 season with $\hat{\theta}_i(0) = 1500$, and following each game, team i 's rating is updated via the formula

$$\hat{\theta}_i(t+1) = \hat{\theta}_i(t) + K \cdot (I(i \text{ beats } j) - \Pr(i \text{ beats } j \text{ at time } t))$$

where $I(i \text{ beats } j)$ is an indicator variable that is 1 if i beats j and -1 if j beats i , and $K = 32$ is a constant that controls how sensitive the ratings are to each game. If team i had a 78% probability of beating team j and wins, their rating will improve by $32 \cdot (1 - 0.78) = 7.04$, while team j 's rating will decrease by the same amount.

We fit Elo ratings using the **elo** package for R. Figure 3 shows the evolution of Elo rating $\hat{\theta}_i(t)$ by franchise for the 12 teams who were active during the 2024 season. The recent poor performance of the Los Angeles Sparks, the 2010 dominance of the Seattle Storm, and the recent rise of the New York Liberty and Indiana Fever are easily visible.

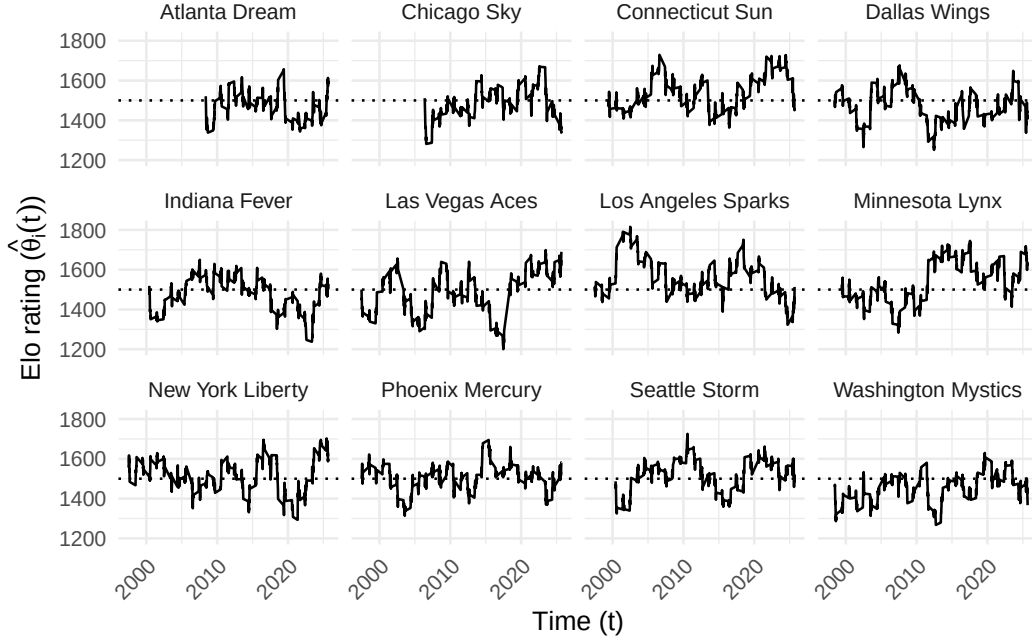


Figure 3: Evolution of team Elo ratings in the WNBA (1997-2025). Only franchises active in 2024 are shown. The recent poor performance of the Los Angeles Sparks, the 2010 dominance of the Seattle Storm, and the recent rise of the New York Liberty and Indiana Fever are easily visible. Note that all franchises begin with a 1500 rating.

Source: [Article Notebook](#)

2.3 Estimating home win probability

The pregame Elo ratings estimated above can then be plugged back into the win probability formula to estimate the home teams's likelihood of winning. We also added a home court advantage equivalent to 60 points (of Elo rating) to our home win probability model, so that:

$$\hat{p}_{ij}(\hat{\theta}_i(t), \hat{\theta}_j(t), t) = \Pr(i \text{ beats } j \text{ at time } t \text{ at home}) = \frac{1}{1 + 10^{\frac{\hat{\theta}_j(t) - (\hat{\theta}_i(t) + 60)}{400}}}.$$

We arrived at 60 points by iteratively calibrating the win probabilities to match the observed frequencies by minimizing the Brier score. While the model was slightly less accurate at the extremes, it fit very well in the middle where most of the games occur (see Table 2).

Source: [Article Notebook](#)

Table 2: Calibration of home win probability model, based on Elo ratings. We note that in the most frequent situations, where the home team was a slight favorite, the observed home win probability is within 3 percentage points of the predicted home probability. However, in those rarer occasions where a home win was extremely likely or unlikely, the model was slightly less accurate.

WNBA Home Win Probability Model Calibration (1997-2025)

10 Equally-spaced Bins

Wpct Bin	Games	Bin Midpoint	Obs Home Wpct	Diff
(0.102,0.189]	61	0.161	0.213	0.052
(0.189,0.275]	251	0.243	0.271	0.028
(0.275,0.361]	422	0.322	0.384	0.062
(0.361,0.446]	691	0.405	0.454	0.050
(0.446,0.532]	970	0.492	0.520	0.029
(0.532,0.618]	1,104	0.577	0.601	0.024
(0.618,0.704]	1,080	0.660	0.661	0.001
(0.704,0.789]	925	0.742	0.706	-0.036
(0.789,0.875]	525	0.825	0.775	-0.050
(0.875,0.962]	135	0.896	0.889	-0.007

3 Methods

Our goal is to model the attendance (y) at a given WNBA game, as a function of several different measures of fan engagement detailed in Section 3.2. We also include several control variables to account for time and city-related effects (see Section 3.3).

Since we know the arena in which the game took place and the capacity of that arena, we developed an alternative response variable- percentage of arena capacity filled (γ).

3.1 Response variables

In Figure 4, we observe the relationship between attendance and estimated home win probability. We add a simple quadratic regression line to reflect the quadratic relationship posited by Coates et al. (2014). In most years, the slope appears to be positive and the curvature concave.

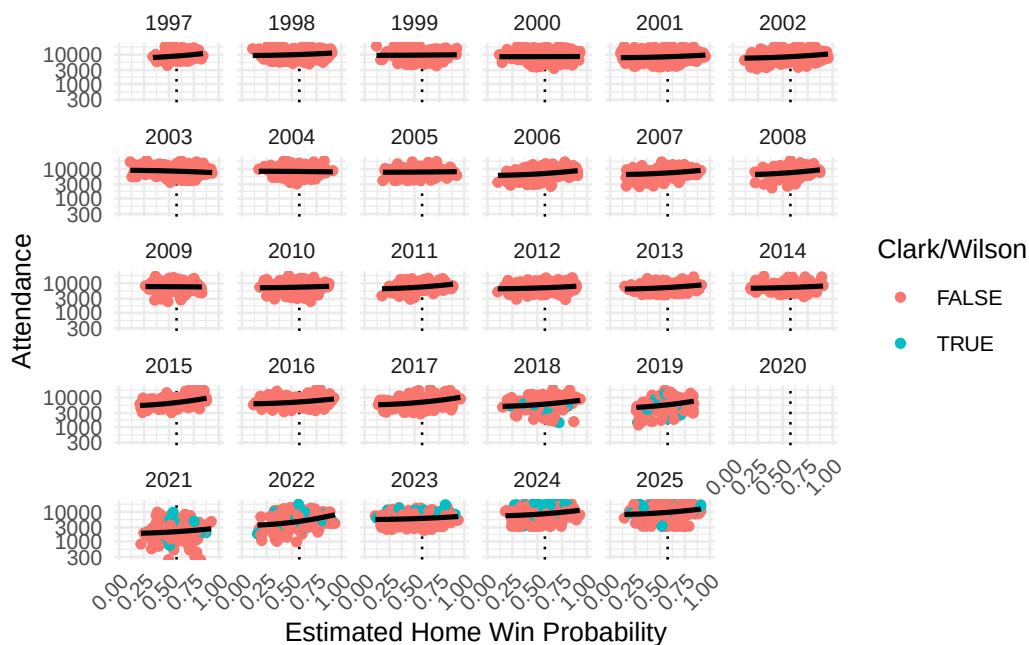


Figure 4: WNBA attendance by season, against estimated home win probability. We note that in most years, the slope of the association appears to be positive and the curvature concave.

Source: [Article Notebook](#)

In Figure 5, we note similar trends in the relationship between the percentage of the arena capacity filled and the estimated home win probability. We also note that since arena capacity cannot exceed 1, there are edge effects created by sellouts. Those sellouts are much more frequent in the last two years of data, and many of the sellouts involve Caitlin Clark.

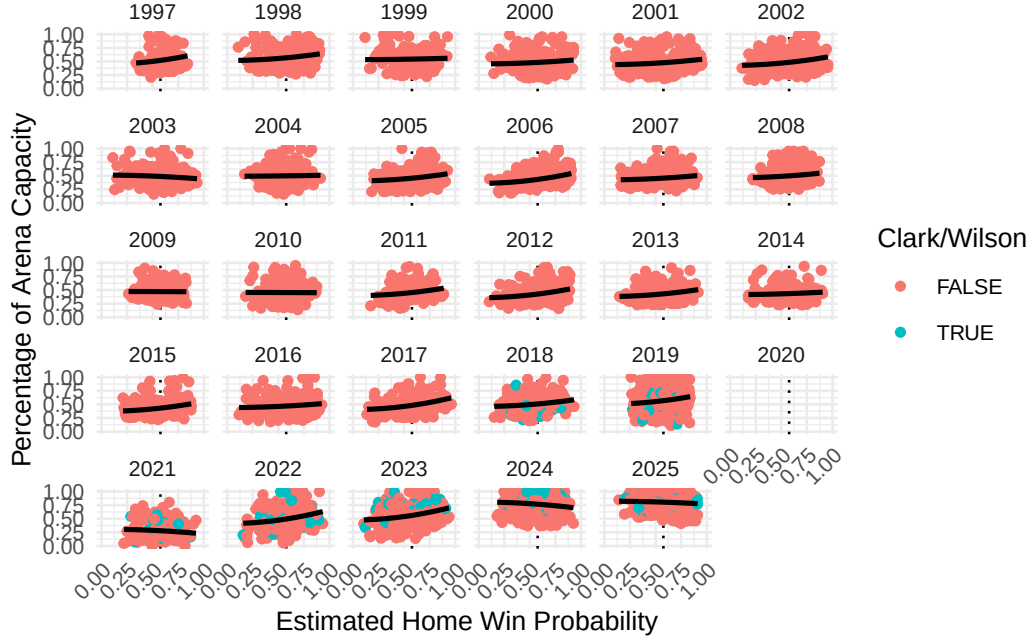


Figure 5: WNBA percentage of arena capacity filled by season, against estimated home win probability. We note the edge effects caused by sellouts, which occur much more frequently in 2024 and 2025.

Source: [Article Notebook](#)

Because the raw attendance figures are not normally distributed, we model the logarithm of attendance. Similarly, because the percentage of arena capacity are bound above by 1 and below by 0, we employ a Tobit regression model.

3.2 Modeling fan engagement

In Section 2.2, we described our use of Elo rating to estimate the strength of the home team ($\hat{\theta}_i(t)$), the strength of the visiting team ($\hat{\theta}_j(t)$) and the home win probability ($\hat{p}_{ij}(t)$) (which is itself a function of the estimated team strengths).

As noted in Section 1, there are multiple notions of what drives fan engagement relevant to the WNBA. Each of these notions can be captured by a simple bivariate function of the corresponding strengths of the teams $f_k(\hat{\theta}_i(t), \hat{\theta}_i(t))$. The first and third have been rescaled so that they will have means close to 0.

- Quality of play: fans may wish to attend a game in person if they expect the quality of play to be high. We model this as the sum of the Elo rating of the two teams ($f_1(\hat{\theta}_i(t), \hat{\theta}_j(t)) = \hat{\theta}_i(t) + \hat{\theta}_j(t)$)
- Uncertainty of outcome hypothesis: fans may wish to attend a game for which the outcome is uncertain. We model this as the absolute difference in the two teams' Elo ratings: ($f_2(\hat{\theta}_i(t), \hat{\theta}_j(t)) = |\hat{\theta}_i(t) - \hat{\theta}_j(t)|$)
- Loss aversion: fans may wish to attend a game to see their team (the home team) win. We model this using the estimated home win probability function above ($f_3(\hat{\theta}_i(t), \hat{\theta}_j(t)) = \hat{p}_{ij}(t)$)

The distribution of attendance with respect to f_1 and f_2 are shown in Figure 6 and Figure 7, respectively, with the corresponding plots for percentage of arena capacity shown in Figure 8 and Figure 9. Note that because our uncertainty of outcome metric is define as the absolute difference in Elo ratings between the two teams. Larger values indicate a less competitive matchup and therefore lower outcome uncertainty. If the uncertainty of outcome hypothesis is true, we would expect a negative relationship between this variable and attendance, as more competitive games should attract larger crowds.

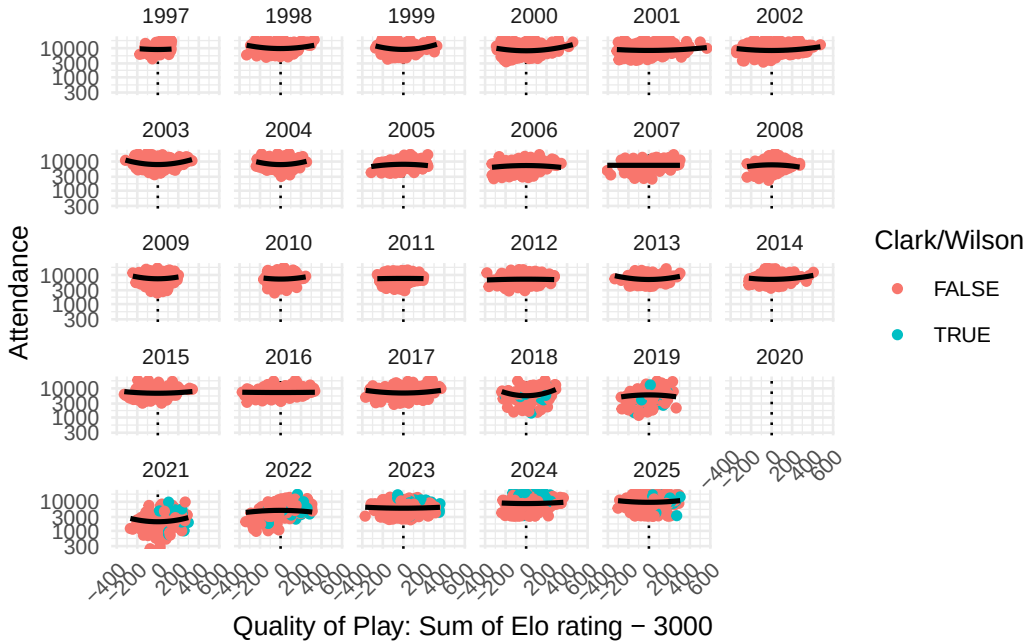


Figure 6: WNBA game attendance by season, against quality of play (game-time sum of Elo ratings - 3000). Higher values indicate matchups between stronger teams.

Source: [Article Notebook](#)

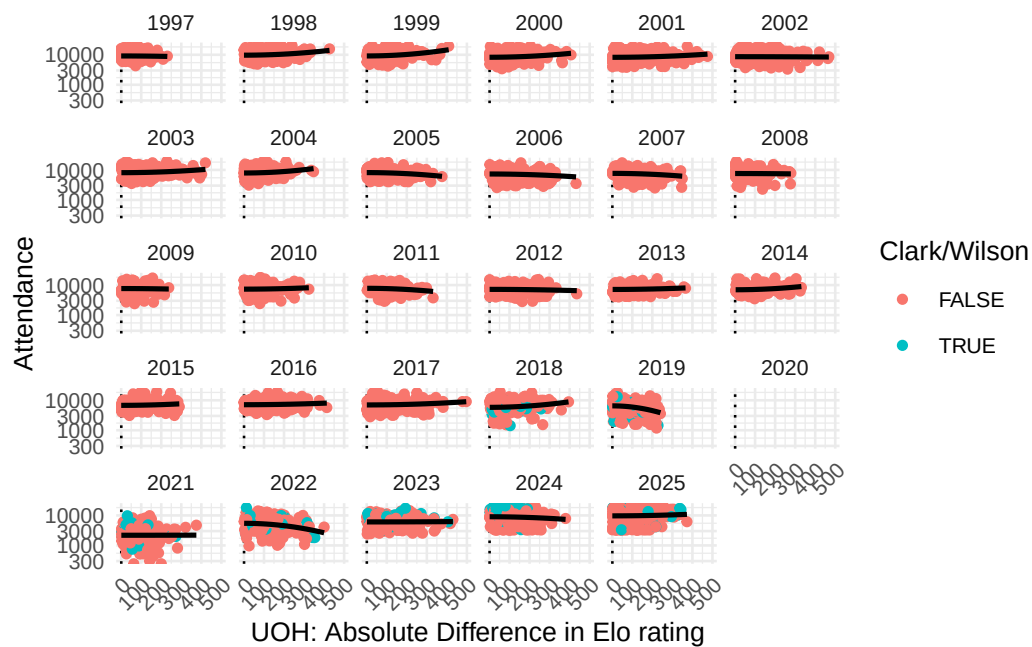


Figure 7: WNBA game attendance by season, against uncertainty of outcome. Games are more competitive when UOH is closer to zero.

Source: [Article Notebook](#)

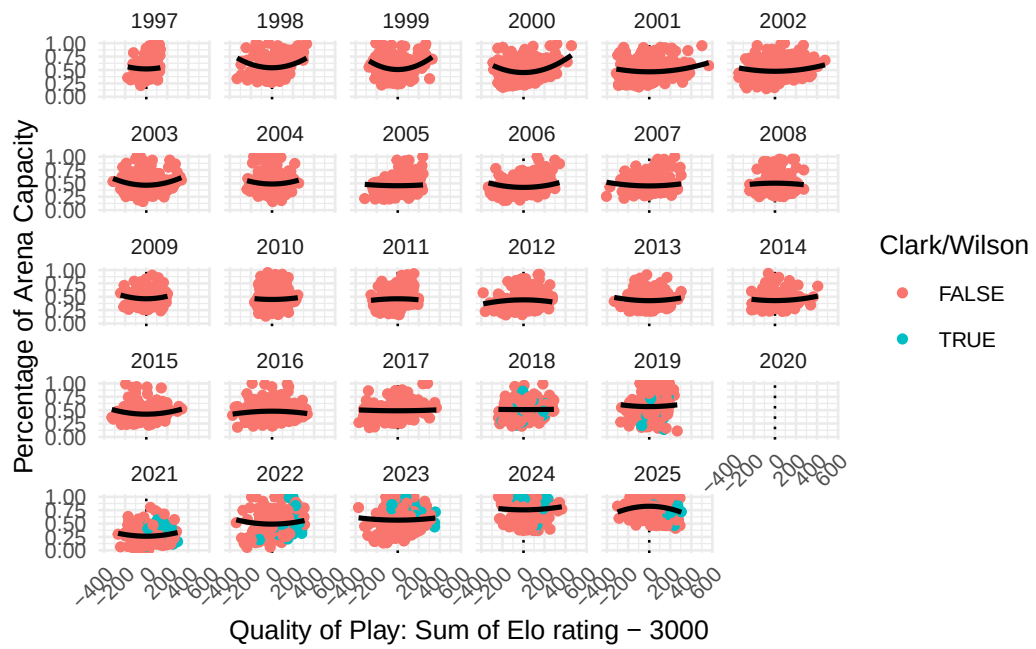


Figure 8: asd

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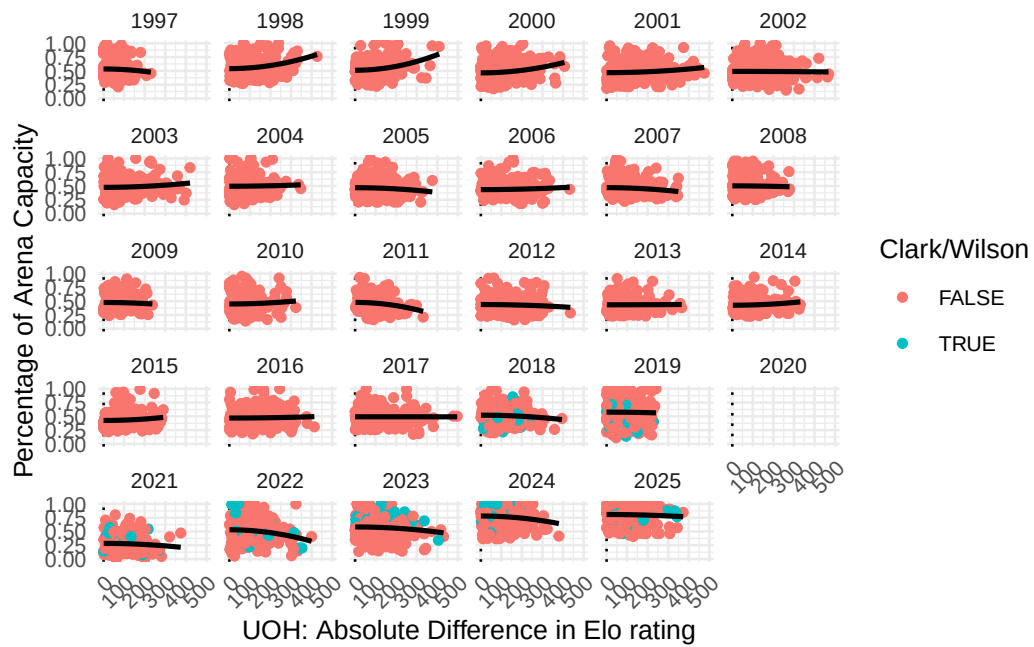


Figure 9: asd

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3.2.1 Rescaling

Figure 10 shows the distribution of our fan engagement variables after rescaling.

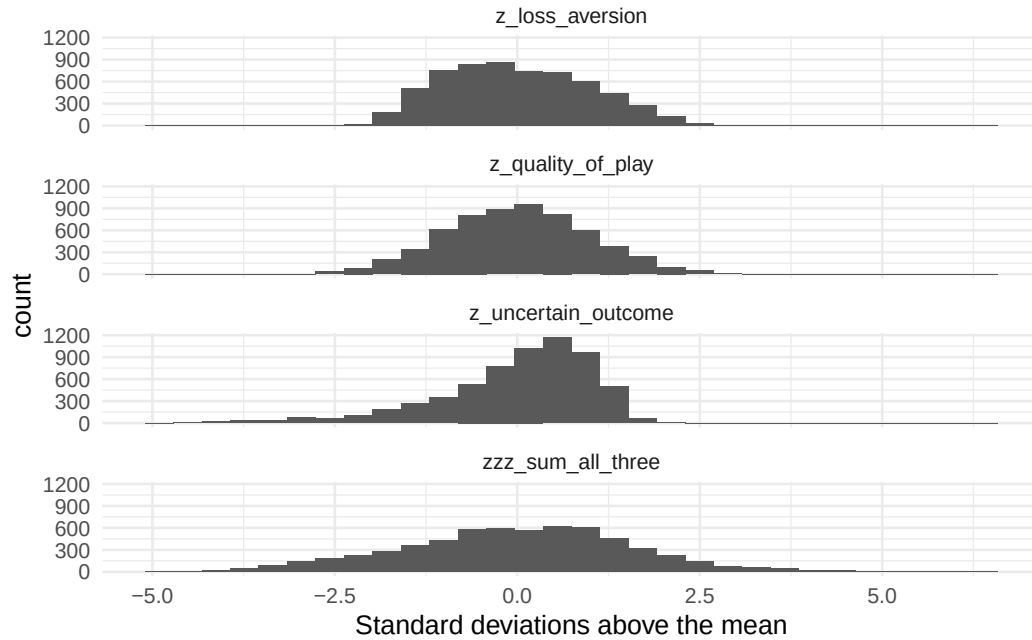


Figure 10: Distribution of fan engagement variables after rescaling.

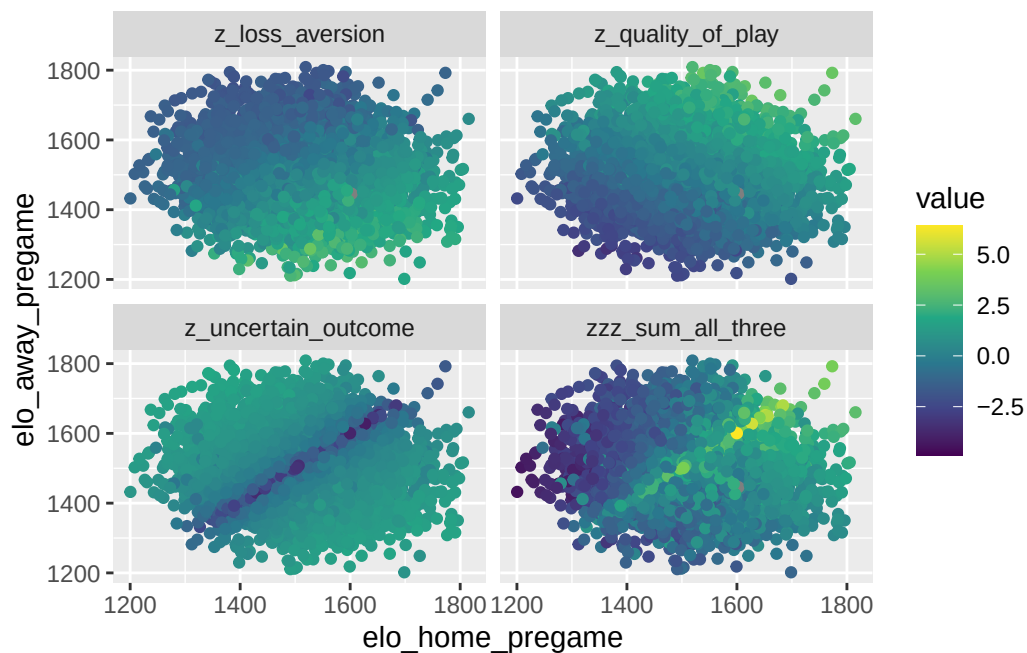
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We note that Table 3 shows that our three fan engagement variables are largely uncorrelated.

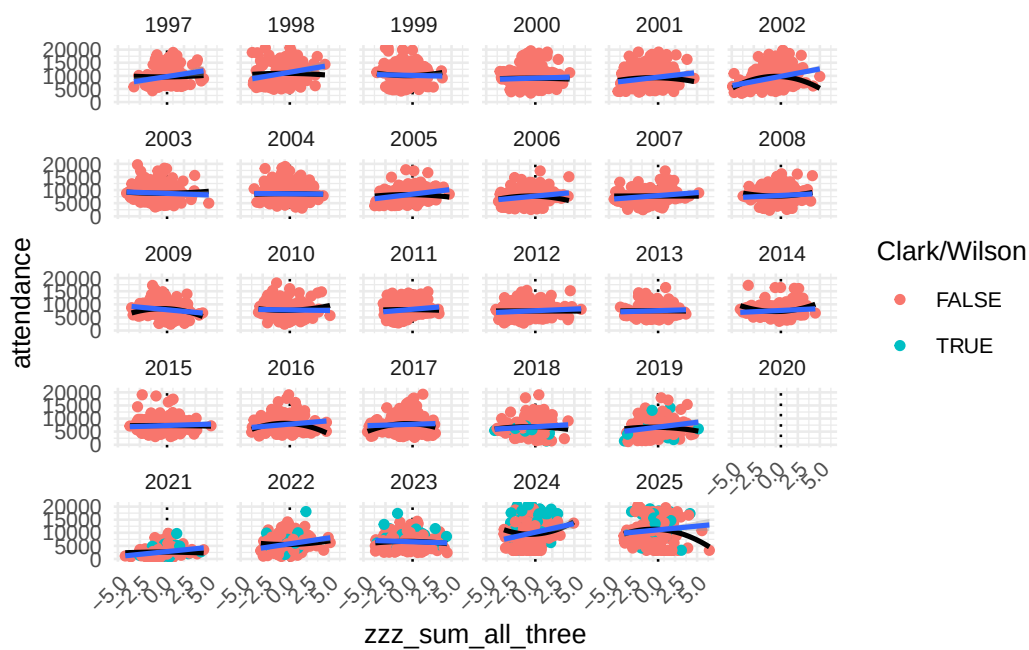
Table 3: Correlations among fan engagment variables.

	z_qual- ity_of_play	z_uncertain_out- come	z_loss_aver- sion	zzz_sum_all_three
z_qual- ity_of_play	1	NA	NA	NA
z_uncertain_out- come	NA	1	NA	NA
z_loss_aversion	NA	NA	1	NA
zzz_sum_all_three	NA	NA	NA	1

Source: [Article Notebook](#)



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3.3 Control variables

As time-related control variables, we include the the season (τ_1), the month of the year (τ_2), and day of the week (τ_3) as fixed effects. To control for city-related effects, we also include the team name (ϕ_1), as well as the median income (ϕ_2 , in thousands of USD) and population (ϕ_3 , in millions) of the home city³. Finally, we include an indicator based on whether Caitlin Clark played more than 0 minutes in the game (χ).

3.4 Interaction

Testing our primary hypotheses requires analyzing how the relationship between attendance and fan engagement has changed over time. To capture these changes, we include an interaction term between our measures of fan engagement (f_k) and season (τ_1). We also considered fitting a period-based model but ultimately decided against this approach. Defining periods by hand would require subjective decisions about the timing of structural breaks and would risk introducing researcher bias into the analysis. Instead, we allow the relationship between attendance and fan engagement to evolve season-by-season.

3.5 Model formulation for attendance

Like Coates et al. (2014), we model the natural logarithm of attendance.

Our general regression model for attendance is:

$$\begin{aligned}
 \ln y &= \beta_0^{(k)} + \underbrace{\beta_1^{(k)} f_k}_{\text{fan engagement}} \\
 &\quad + \underbrace{\beta_2^{(k)} \tau_1 + \beta_3^{(k)} \tau_2 + \beta_4^{(k)} \tau_3}_{\text{time-related}} + \underbrace{\beta_5^{(k)} \phi_1 + \beta_6^{(k)} \phi_2 + \beta_7^{(k)} \phi_3}_{\text{city-related}} \\
 &\quad + \underbrace{\beta_8^{(k)} \chi}_{\text{Caitlin Clark}} + \underbrace{\beta_9^{(k)} f_k \tau_1}_{\text{interaction}} + \epsilon \\
 &= \mathbf{X}^{(k)} + \epsilon,
 \end{aligned}$$

where $k \in \{1, 2, 3\}$, and $\beta_j^{(k)} \in \mathbb{R}$ for $j = \{0, 1, 6, 7, 8\}$, while the other $\beta_j^{(k)}$ coefficients are vectors of length equal to the number of factor levels in their corresponding categorical explanatory variable. Most notably $\beta_2^{(k)}$ and $\beta_9^{(k)}$ have 29 values (corresponding to each season after 1997)

³The data for the population and the median income of cities comes from the American Community Survey, access to which was provided in R by the **tidycensus** package. Because the population and median income statistics were available at irregular time periods (due to the COVID-19 pandemic and other factors), we chose to use the median of the population and median income across all of the available years of data as a constant value. While this does mean that the population growth in cities like Phoenix or Las Vegas is ignored, we doubt this will meaningfully affect our results, especially since the team fixed effect should capture some of that variability.

and $\beta_5^{(k)}$ has 19 values (corresponding to each team). Only current WNBA franchises are included in the regression model. We also excluded all data from the 2020 and 2021 seasons, since there were no fans allowed in all or parts of both seasons.

3.6 Model formulation for percentage of arena capacity

We fit the same set of models with γ as the response, but because our response variable γ is constrained between 0 and 1 and has significant mass at 1 exactly, we use the Papke-Wooldridge fractional response model (Papke and Wooldridge 1996). This is equivalent to fitting a generalized linear model with a logit link function and applying HC1 sandwich estimators (MacKinnon and White 1985) for the covariance matrix, which we accomplish using the **sandwich** package in R.

$$\mathbb{E}(\gamma|\mathbf{X}) = G(\mathbf{X}^{(k)}),$$

where $G(z) = \frac{e^z}{1+e^z}$ is the (vectorized) logistic function, $\mathbf{X}$ is as above, and $^{(k)}$ has the same structure as $^{(k)}$ defined above.

4 Results

Source: [Article Notebook](#)

Source: [Article Notebook](#)

4.1 Attendance

In keeping with recent guidance from Wasserstein and Lazar (2016) and Wasserstein et al. (2019), and with a sample size of over 5000 observations and dozens of coefficients, we are less interested in the p-values shown in Table 4, and focus our attention on the magnitude and practical impact of the fitted coefficients. (All variables except for population achieve “statistical significance” at the 5%-level, as shown in Table 8). Residual analysis for the attendance models are shown in Section 6.2.

4.1.1 Baseline effects

Table 4: Fitted coefficients for scalar terms in models for the natural logarithm of attendance.

Model	Term	Estimate	Std Error	Statistic	P Value	Conf Low	Conf High
f1	(Intercept)	9.202	0.081	113.091	0.000	9.042	9.361
f1	I(Z Loss Aversion)	-0.081	0.048	-1.690	0.091	-0.175	0.013

Table 4: Fitted coefficients for scalar terms in models for the natural logarithm of attendance.

Model	Term	Esti- mate	Std Error	Statistic	P Value	Conf Low	Conf High
f1	I(Z Quality Of Play)	0.003	0.059	0.058	0.954	-0.112	0.119
f1	I(Z Uncertain Outcome)	-0.053	0.022	-2.407	0.016	-0.097	-0.010
f1	Is Sellouttrue	0.315	0.027	11.616	0.000	0.261	0.368
f1	Median Income	NA	NA	NA	NA	NA	NA
f1	Median Pop	NA	NA	NA	NA	NA	NA
f1	P: A'ja Wilson	0.061	0.024	2.573	0.010	0.014	0.107
f1	P: Caitlin Clark	0.460	0.042	10.825	0.000	0.376	0.543

Source: [Article Notebook](#)

Table 4 reports estimates from the baseline interaction. Because each measure of fan engagement is interacted with season, the reported coefficients for quality of play, outcome uncertainty, and loss aversion represent their estimated effects in the reference season (1997). The interaction terms, discussed below, capture how these relationships evolve over time. In the reference season, the evidence for the quality of play (f_1) and uncertainty of outcome (f_2) hypotheses is weak, with those coefficients barely moving the needle. For the loss aversion hypothesis, a 10 percentage point increase in the probability of the home team winning is associated with a modest 4.31% increase in the expected attendance, after controlling for our other factors.

Among the control variables, games featuring Caitlin Clark were associated with significantly higher attendance, while A'ja Wilson's presence was associated with a smaller but still significant increase in attendance. Even for the smallest coefficient in Table 4, the expected attendance nearly doubles (i.e., is multiplied by $\exp(0.623) = 1.58$) when Clark plays in the game, and this is *after* controlling for the Indiana Fever team effect, the level of fan engagement, and the season effects! The corresponding change in attendance for A'ja Wilson—by most accounts a better player than Clark—is a multiple of 1.06.

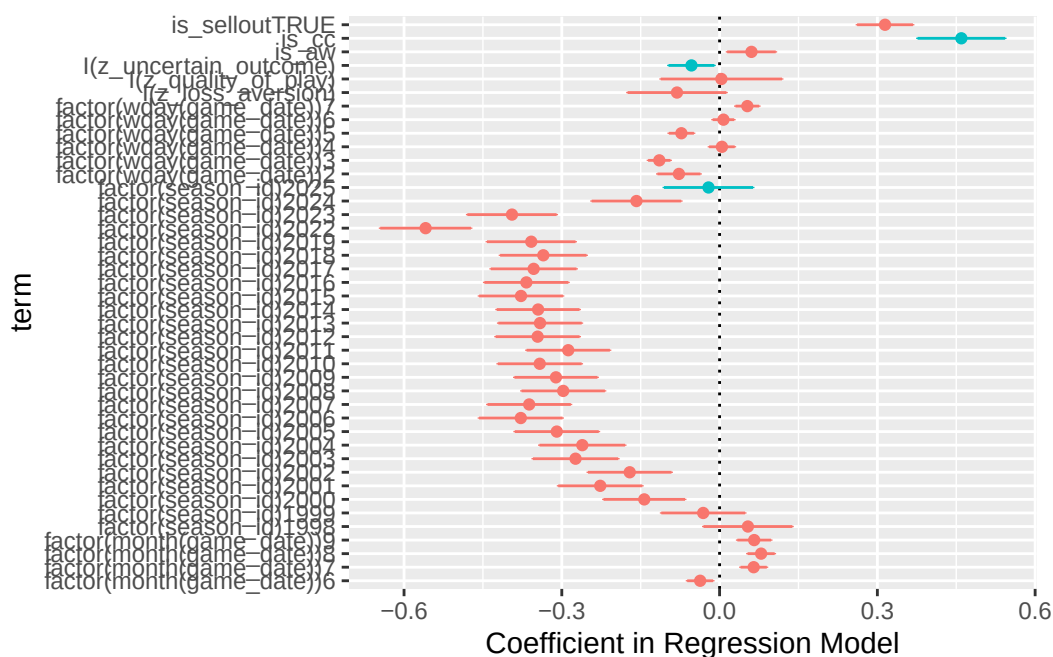


Figure 11: Relative magnitudes of coefficients in attendance model. Coefficients of interest are highlighted. Note the relatively large size of the coefficient associated with Caitlin Clark.

Source: [Article Notebook](#)

Table 5 characterizes the model’s fit. Our model explains 70.2 of the variance in game attendance, and the small difference between r^2 and $adj.r^2$ suggests that the complexity of our model is justified.

Table 5: Model fit

	model	r.squared	adj.r.squared	sigma	statis- tic	p.value	df	log- Lik	AIC	BIC	de- viance	df.resid- ual	nobs
f1		0.702	0.691	0.233	66.734	0	172	274.025	-	935.879	265.618	4883	5056
								200.05					

Source: [Article Notebook](#)

Model	Residual df	Residual SS	Df added	Δ SS	F- statistic	p- value
Baseline (no season-varying slopes)	4961	275.43	NA	NA	NA	NA
Season-varying slopes	4883	265.62	78	9.814	2.313	0

Source: [Article Notebook](#)

Table 8 compares a model fit with constant slope coefficients to our model which allows each effect to vary by season. The interactions introduce 78 additional terms corresponding to season specific slopes to our model. The nested F-test rejects the null hypothesis that season-specific interactions are equal to zero, indicating that the effects of our fan engagement measures differ significantly across seasons.

4.1.2 Annual effects

Figure 12 shows that after controlling for the Caitlin Clark effect, expected attendance is only just returning its 1997 baseline in the most recent season. This is a mixed blessing. On the one hand, even after controlling for Clark, league expected attendance has fully recovered from the COVID-19 pandemic and the malaise of the early 2000s and 2010s. On the other hand, it seems unlikely that the league would have achieved record high attendance without Clark.

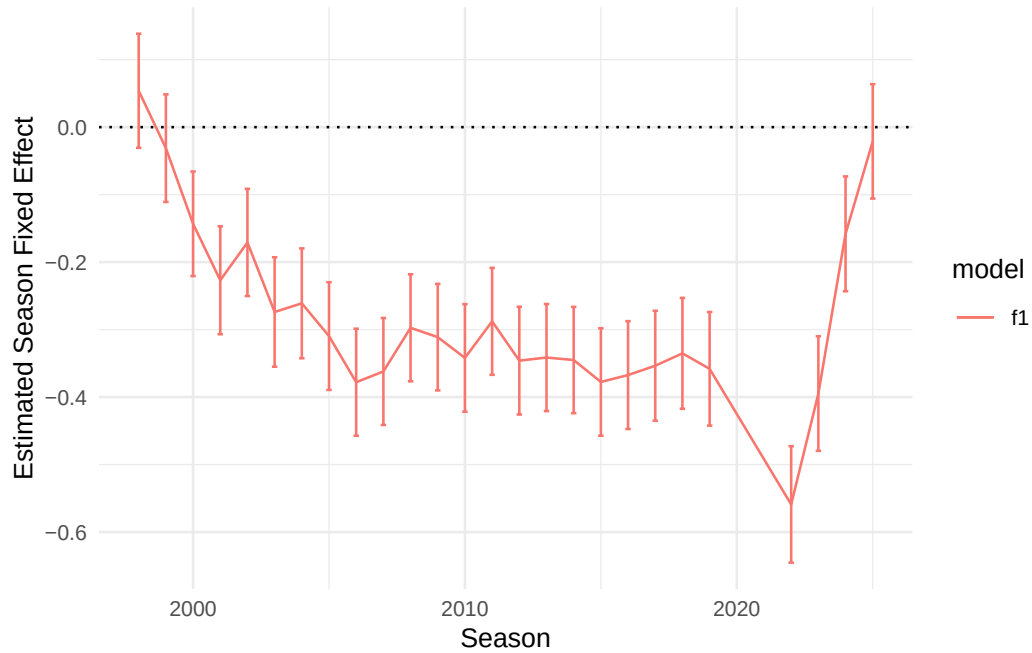


Figure 12: Estimated season-level fixed effects for natural logarithm of attendance models.

Source: [Article Notebook](#)

Figure 13 shows the season-by-season slopes on the interaction terms for our three hypotheses, relative to the 1997 reference season. Where the seasonal fixed effects convey broad attendance trends, the interaction term slopes identify trends in how fan behavior is driven by game outcomes. Where general attendance dropped off after the opening season, the extent to which fan behavior is influenced by game outcome seems to improve past the first season. In particular, the interaction coefficients on our loss aversion variable are positive and generally increasing, suggesting that the attendance response to home win probability has strengthened over the league's history. The importance of outcome uncertainty and quality of play on attendance appears to be weaker and not substantially improved in recent years.

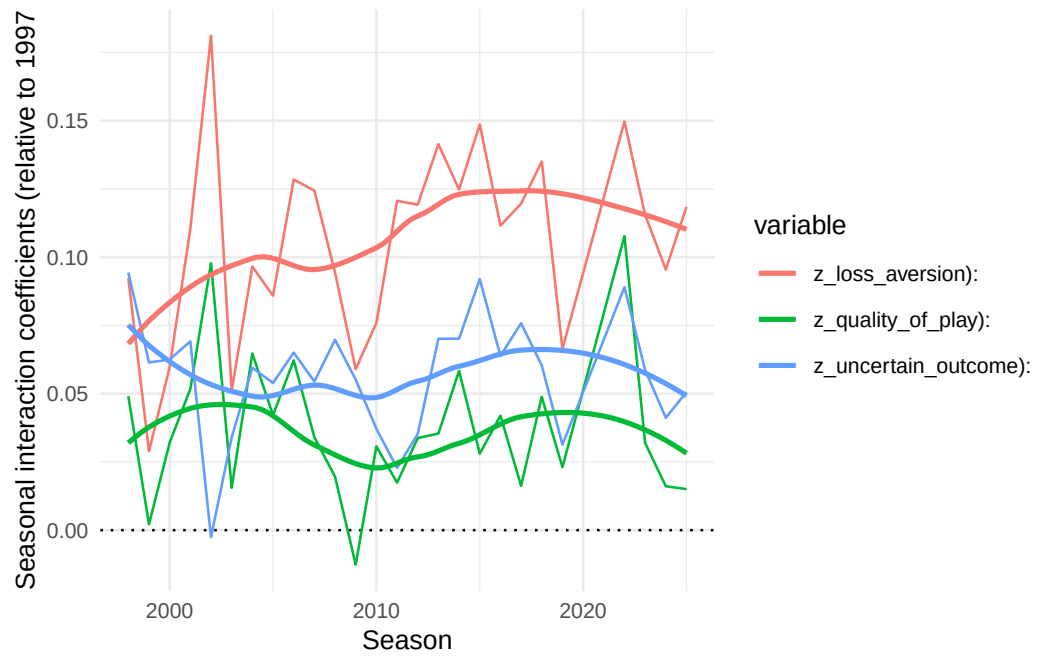


Figure 13: Estimated season-level changes in slopes for natural logarithm of attendance models.

Source: [Article Notebook](#)

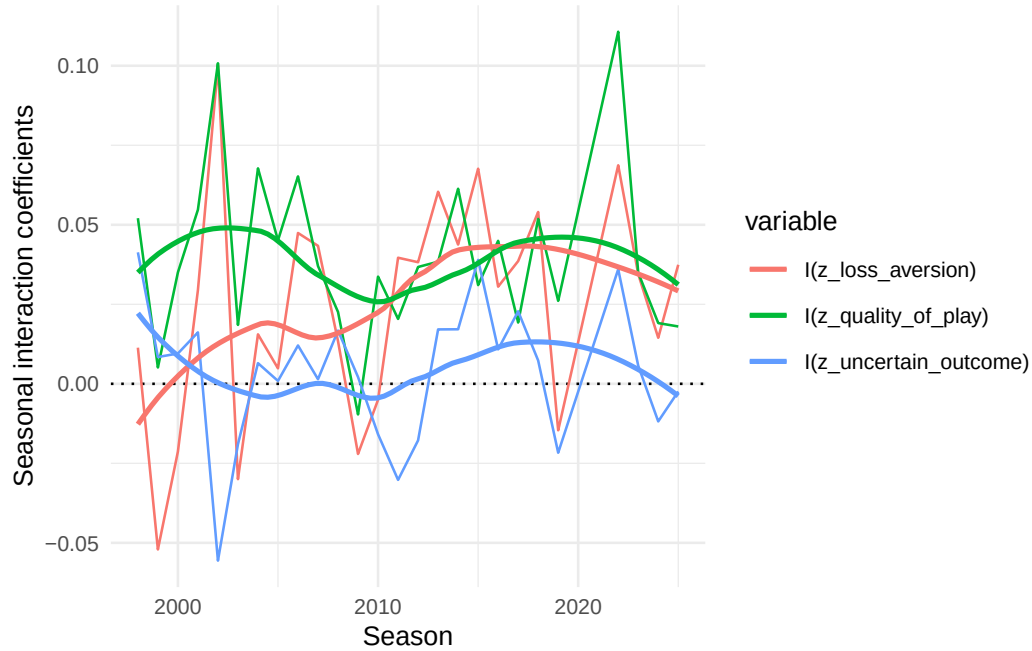


Figure 14: Estimated season level slopes for attendance models

Source: [Article Notebook](#)

4.2 Percentage of Arena Capacity

Because our response variable γ is constrained between 0 and 1, but has significant mass at 1 exactly, we fit a Papke-Wooldridge fractional response model (Papke and Wooldridge 1996) with HC1 sandwich estimators for the covariance matrix.

Source: [Article Notebook](#)

4.2.1 Main effects

Table 6 reports the coefficients from the percentage of arena capacity models.

Table 6: Table of coefficients from percentage of arena capacity models.

Model	Term	Esti- mate	Std Error	Statistic	P Value	Conf Low	Conf High
f1	(Intercept)	2.197	0.218	10.091	0.000	1.770	2.624
f1	I(Z Loss Aversion)	-0.170	0.117	-1.444	0.149	-0.400	0.061

Table 6: Table of coefficients from percentage of arena capacity models.

Model	Term	Estimate	Std Error	Statistic	P Value	Conf Low	Conf High
f1	I(Z Quality Of Play)	0.111	0.159	0.701	0.483	-0.200	0.423
f1	I(Z Uncertain Outcome)	-0.123	0.057	-2.168	0.030	-0.234	-0.012
f1	Is Sellouttrue	2.888	0.118	24.384	0.000	2.656	3.121
f1	P: A'ja Wilson	0.175	0.054	3.257	0.001	0.070	0.280
f1	P: Caitlin Clark	1.613	0.212	7.599	0.000	1.197	2.030

Source: [Article Notebook](#)

Figure 15 shows that...

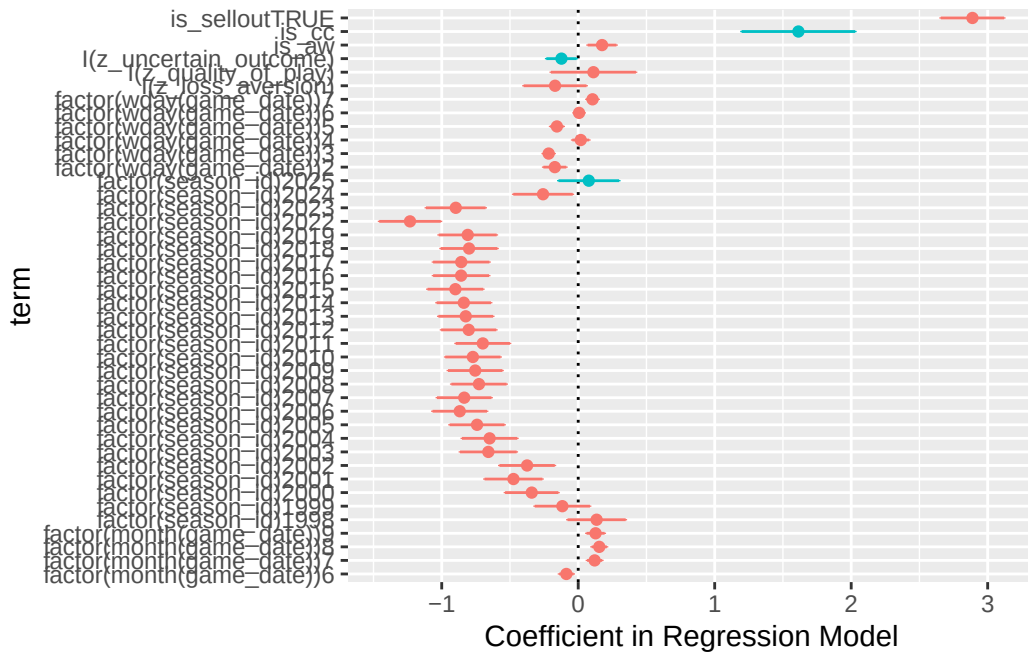


Figure 15: Relative magnitudes of coefficients in attendance model. Coefficients of interest are highlighted. Note the relatively large size of the coefficient associated with Caitlin Clark.

Source: [Article Notebook](#)

Table 7 shows the model fit metrics for the percentage of arena capacity models.

Table 7: Comparison of percentage of arena capacity models.

model	null.de- viance	df.null	logLik	AIC	BIC	de- viance	df.residual	nobs
f1	860.2571	5055	- 2705.892	5757.784	6887.185	266.5421	4883	5056

Source: [Article Notebook](#)

Table 9 shows the analysis of variance for the percentage of arena capacity models.

4.2.2 Annual effects

Figure 16 displays the estimated effects of our fan engagement measures on attendance relative to the 1997 reference season. Similar to the overall attendance trends, these effects generally remained below their 1997 levels for much of the league's history and only returned to comparable levels recently. In the most recent years of measurement, however, we see the league exceeding previously record-holding levels.

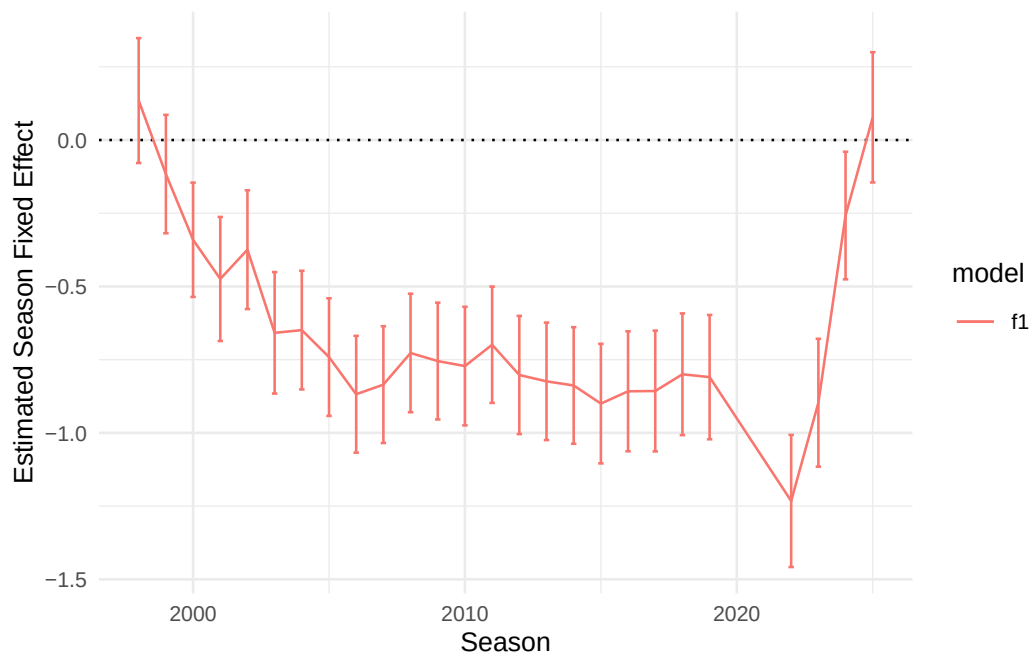


Figure 16: Estimated season-level fixed effects for percentage of arena capacity models.

Source: [Article Notebook](#)

Figure 17 shows that...??

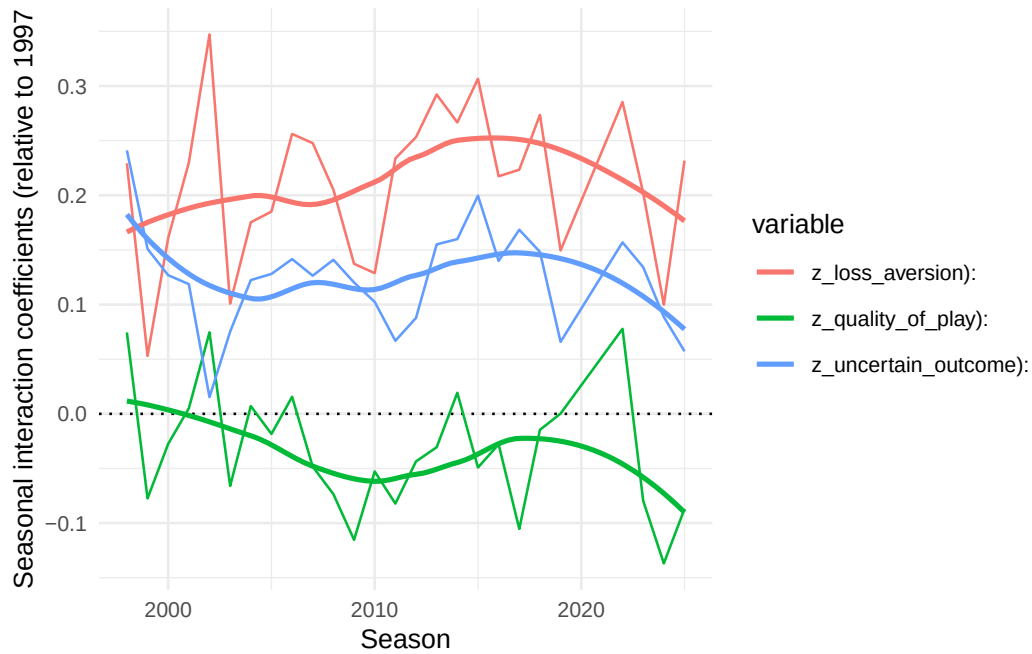


Figure 17: Estimated season-level changes in slopes for percentage of arena capacity models.

Source: [Article Notebook](#)

5 Conclusion

5.1 Limitations

Because the percentage of arena capacity data is right-censored, we considered a Tobit regression model (McDonald and Moffitt 1980), but our residuals violated the normality and homoskedasticity conditions. We also considered fitting a Beta regression model with a logit link function, and converting the fixed effects for seasons to random effects, but those models didn't converge.

5.2 Future work

5.3 Discussion

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6 Appendix

6.1 ANOVA tables

Table 8: Analysis of variance for terms in models for the natural logarithm of attendance.

Model	Term	Df	Sumsq	Meansq	Statistic	P Value
f1	I(z_loss_aversion)	1	2.981	2.981	54.793	0.000
f1	I(z_quality_of_play)	1	5.733	5.733	105.392	0.000
f1	I(z_uncertain_outcome)	1	0.003	0.003	0.051	0.822
f1	is_aw	1	0.511	0.511	9.399	0.002
f1	is_cc	1	13.501	13.501	248.198	0.000
f1	is_sellout	1	8.349	8.349	153.486	0.000
f1	factor(month(game_date))	4	12.297	3.074	56.518	0.000
f1	factor(wday(game_date))	6	14.302	2.384	43.820	0.000
f1	I(z_loss_aversion):factor(season_id)	26	4.638	0.178	3.280	0.000
f1	I(z_quality_of_play):factor(season_id)	26	2.373	0.091	1.678	0.017
f1	I(z_uncertain_outcome):factor(season_id)	26	2.803	0.108	1.982	0.002
f1	factor(season_id)	26	154.156	5.929	108.997	0.000
f1	factor(arena)	52	402.731	7.745	142.378	0.000

Source: [Article Notebook](#)

Table 9: Analysis of variance for terms in models for percentage of arena capacity.

Model	Term	Df	De- viance	Df Residual	Residual Deviance	P Value
f1	median_income	0	0.000	4964	341.054	NA
f1	median_pop	0	0.000	4964	341.054	NA
f1	I(z_loss_aversion)	1	2.816	5052	852.070	0.093
f1	I(z_quality_of_play)	1	5.353	5054	854.904	0.021
f1	I(z_uncertain_outcome)	1	0.017	5053	854.887	0.895
f1	is_aw	1	0.835	4962	317.728	0.361
f1	is_cc	1	22.492	4963	318.563	0.000
f1	is_sellout	1	41.696	4961	276.032	0.000
f1	factor(month(game_date))	4	11.663	5016	611.101	0.020
f1	factor(wday(game_date))	6	15.055	5020	622.764	0.020
f1	I(z_loss_aversion):factor(sea- son_id)	26	4.302	4883	266.542	1.000
f1	I(z_quality_of_play):factor(sea- son_id)	26	2.500	4935	273.532	1.000
f1	I(z_uncertain_outcome):fac- tor(season_id)	26	2.688	4909	270.844	1.000
f1	factor(season_id)	26	214.251	5026	637.819	0.000
f1	factor(arena)	52	270.047	4964	341.054	0.000

Source: [Article Notebook](#)

6.2 Residual analysis of attendance models

6.3 Tobit modeling

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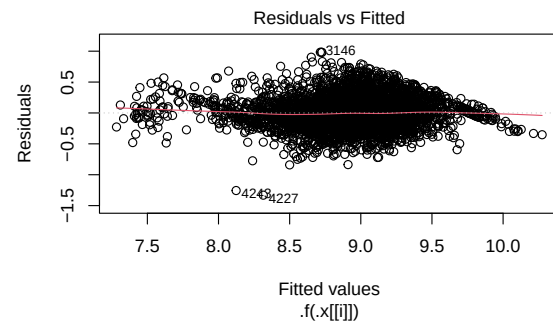


Figure 18: Residual plots for models.

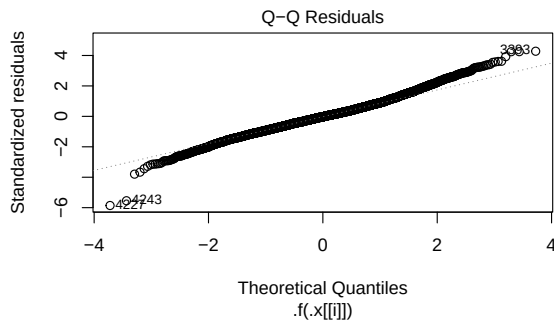


Figure 19: Residual plots for models.

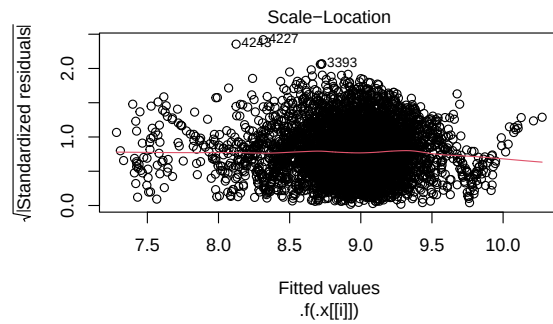


Figure 20: Residual plots for models.

Source: [Article Notebook](#)

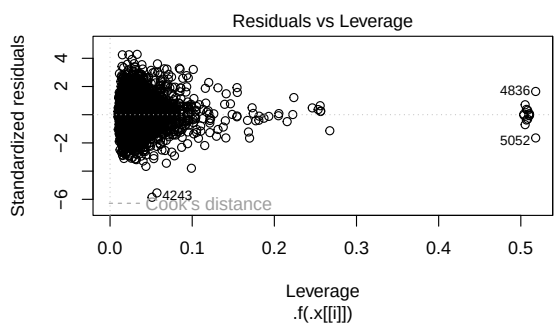


Figure 21: Residual plots for models.